A Multilevel-Guided Curriculum Domain Adaptation Approach to Semantic Segmentation for High-Resolution Remote Sensing Images

Zhihao Xi, Xiangyu He, Yu Meng, Anzhi Yue, Jingbo Chen, Yupeng Deng, and Jiansheng Chen

***Abstract*— The semantic segmentation of high-resolution (HR) remote sensing images (RSIs) has been extensively researched in various applications. However, segmentation networks are prone to significant performance degradation on unlabeled data due to domain shift, such as data distribution shifts arising from distinct geographic locations. To address this issue, we propose a multilevel-guided curriculum domain adaptation (MuGCDA) approach for joint samplewise, categorywise, and pixelwise tasks, which facilitates the final fine-grained segmentation task by guiding the target domain to acquire samplewise and categorywise domain-robust properties. Concretely, at the sample level, we formulate a sample spatial relationship consistency guidance (SSCG) loss that guides the target domain to acquire similar sample spatial relationship properties to the source domain. At the category level, we propose a category layout structure consistency guidance (CLCG) module that guides the target domain to acquire consistent layout properties. At the pixel level, we design an adaptive hierarchical pseudolabel weight setting (AHPWS) method with a self-training (ST) paradigm to reduce the effect of label noise while improving the quality of the generated pseudolabels. Furthermore, to improve the stability of the training process, we use a momentum network (MN) as the teacher network to obtain the property knowledge and pseudolabels, and then guide the whole domain transfer process of the segmentation network, which acts as the student network. Extensive comparison and ablation experiments are conducted in several cross-space and cross-spectral scenes, and the results show that our method achieves significant performance improvements in cross-domain scenes for HR RSIs.**

***Index Terms*— Domain adaptation (DA), high-resolution (HR) remote sensing, multilevel-guided curriculum learning, semantic segmentation.**

I. Introduction

With the rapid development of remote sensing technology, high-resolution (HR) remote sensing images (RSIs) have become increasingly varied and abundant, and the acquisition quality has greatly improved, demonstrating the application potential of HR RSIs. However, the complex spatial information contained in HR RSIs increases the technical difficulty of interpretation [1]. Semantic segmentation, which is a classification task that involves assigning specific semantic labels to each pixel in input image, plays an important role in information extraction and intelligent interpretation of HR RSIs [2], [3], [4].

Recently, the performance of semantic segmentation networks has significantly improved [5], [6]. However, these fully supervised learning approaches are based on sufficient labeled data, and acquiring these data requires considerable labor and time costs. Moreover, their performance degrades sharply if the training (source domain) and test (target domain) data suffer significant domain discrepancies, where the images follow different data distributions. Spatial locations and spectral variations are the fundamental features of RSIs, and the condition that the images must have identical distributions is difficult to satisfy. The source domain data might include historical RSIs in a certain region, for which we can obtain the corresponding ground truth, whereas the target domain data may come from a different region, for which we cannot acquire the corresponding ground truth. Due to different geographic locations and atmospheric radiation conditions, the local object features in RSIs may cause shifts, and it is often difficult to achieve satisfactory segmentation results if the model is trained only on the source domain data and applied directly to the target domain data.

An effective solution to this problem is the unsupervised domain adaptation (DA) approach, which aims to reduce the interdomain discrepancies and improve the generalizability of the model by transferring the model knowledge learned on the labeled source domain data to the unlabeled target domain data. Recently, DA methods based on adversarial training (AT) have been intensively studied [7], [8], [9] and have attracted considerable attention in the field of remote sensing [10], [11], [12], [13], [14], [15]. These approaches typically attempt to alleviate the domain discrepancy problem by aligning the feature distributions of the source and target domains to obtain a domain-invariant feature space. In addition, domain transfer for input images has been widely explored in remote sensing [16], [17], [18], [19], [20]. This approach leverages the adversarial-style transfer technique to generate fake target domain images. The semantic information of the transferred fake target domain images is identical

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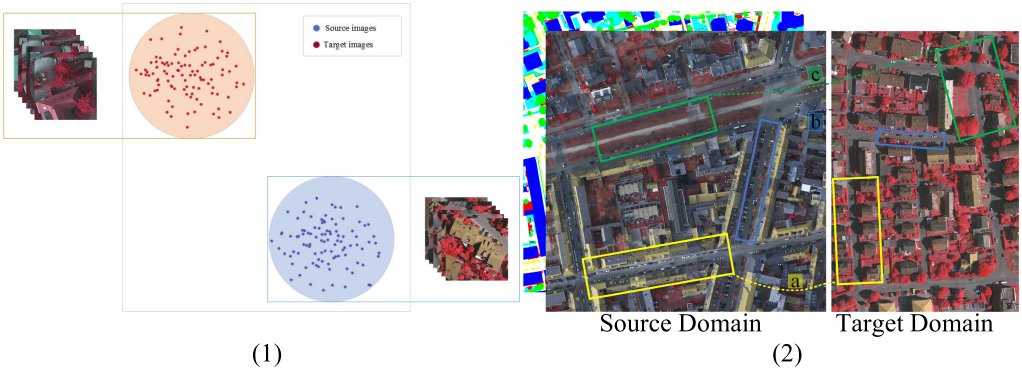


Fig. 1. Illustration of the data characteristics of HR RSIs. (1) Dimensionality reduction visualization of datasets. Similar sample space relationship properties between the source and target domains: different datasets constitute their own clusters, while intradomain samples present similar spatial aggregation properties. (2) Similar category spatial layout properties between the source and target domains: (a) Buildings exist on both sides of the road; (b) vehicles drive or park on the road; and (c) accompanying trees and grasses.

to that of the source images but follows a similar data distribution to the target domain images. Self-training (ST) methods, which are actually semisupervised approaches, have also been investigated for domain-adaptive semantic segmentation [21], [22], [23]. These techniques iteratively generate pseudolabels based on the target domain and then participate in the next round of training with the labeled source domain data, thus progressively aligning the feature spaces of the source and target domains. Essentially, ST-based DA methods have the same goal as adversarial-based feature alignment methods in that both the approaches attempt to find a feature space that can match the distributions of the source and target domains. However, many AT methods focus on exploring sophisticated, high-dimensional output spaces. Although this approach can achieve more fine-grained global and local feature space alignment, it also increases the complexity and difficulty of the AT process [7], [24], [25]. Furthermore, this approach may neglect some simple but crucial structural consistency properties between the source and target domains, such as similar layout structures in the output space [26]. Image transfer techniques also require sophisticated optimization processes for generative adversarial networks (GANs) and are prone to generating somewhat unnatural objects that are inconsistent with the semantic information of the original image. In contrast, ST-based approach provides a simpler and more elegant way to solve the domain discrepancy problem, but the alignment direction of its features is irregular, so the stability of the model and the effectiveness of domain transfer cannot be guaranteed.

Considering these limitations and practical remote sensing applications, we focus on the cross-space location domain discrepancy problem. Intuitively, we can obtain two characteristics by observing the sample data of HR RSIs. First, because of memory constraints, the input samples of the model are often cropped to a fixed size from the original oversized images, which implies the existence of latent spatial relationships between the samples and that these relationships are preserved in the structure of the data embedding space. As shown in Fig. 1(a), we use the t-SNE [27] method to visualize the feature distributions in the source and target domains. Different datasets generate individual clusters, while intradomain samples present similar spatial aggregation properties. Although data distribution shifts between domains can occur due to the variations in the spatial location and atmospheric radiation conditions, these influences are holistic for intradomain samples, which means that the spatial relationships of intradomain samples are essentially independent of these factors. On the other hand, Fig. 1(b) shows different domains have similar category spatial layout properties that are less affected by domain discrepancies [26], [28]. For instance, buildings exist on both sides of roads, vehicles drive or park on roads, and accompanying trees and grasses in urban remote sensing scenes. These similar category layouts directly respond to the output spatial structure.

Based on the above observations, we consider curriculum learning idea and propose a multilevel-guided curriculum DA (MuGCDA) approach for semantic segmentation of HR RSIs. We combine online ST methods and curriculum learning strategies to develop a simple feature alignment method without adversarial attributes. Specifically, instead of directly conducting complex fine-grained alignment on the target domain features, we constrain the model to focus on domain-robust properties that are less affected by domain discrepancies, including mutual spatial relationship consistency properties at the sample level and layout structure consistency properties at the category level. It is a relatively simple but global and region-guided task for the model to learn these prior properties of the target domain, which can guide the semantic segmentation task based on these properties and acquire a rough alignment direction during the transfer process. In addition, we design an adaptive hierarchical pseudolabel weight setting (AHPWS) method. Instead of selecting or neglecting the corresponding labels based on confidence thresholds, we adaptively set the pseudolabel weights in a hierarchical manner to flexibly balance label noise with hard samples and hard classes that tend to have low confidence values.

The main contributions can be summarized as follows.

1) We propose a novel MuGCDA network, which constructs easy-to-hard multilevel tasks to promote DA performance. We first design a sample space relation consistency guidance (SSCG) mechanism and a category layout structure consistency guidance (CLCG) mechanism to achieve simple guidance alignment tasks. Then, we use the ST method for fine-grained domain transfer.

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2) We design an AHPWS method, which adaptively stratifies the pseudolabel weight values based on the output structure of the samples. This method effectively reduces the impact of label noise, allowing more hard classes and hard samples to participate in the training process.
3) We propose a generalized approach that does not require modifying the structures of the existing prevalent semantic networks. Furthermore, the results of experiments demonstrate the effectiveness of the proposed method.

II. Related Work

A. Semantic Segmentation in Remote Sensing

Fully convolutional neural networks (FCNs) are landmark semantic segmentation methods that assign labels to each pixel of an input image in an end-to-end manner. FCN-based methods have been extensively studied in the field of computer vision, including the DeepLab series of networks, which can capture multiscale perceptual field information [29], [30], [31], U-Net, which has a classic U-shaped network structure [32], the dual-attention network (DANet), which has spatial position and channel attention mechanisms [33], and the HR network (HRNet), which has an overall structure that maintains HR feature information [34]. Many studies have been conducted on RSIs in the field of intelligent interpretation and have achieved significant progress [35], [36]. These methods have achieved improved semantic segmentation performance on RSIs by focusing on different aspects, including global and local contextual information [37], [38], edge optimization losses [2], [39], multilevel feature fusion to reduce the spatial information loss [35], and self-attention mechanisms [40]. Most semantic segmentation networks are designed based on the encoder–decoder architectures, with the encoder determining the feature extraction capability of the network. Some prevalent encoders based on convolutional neural networks (CNNs) include VGGNet [41], ResNet [42], EfficientNet [43], and HRNet. Furthermore, transformer-based architectures have recently been widely studied and applied in the field of computer vision, achieving excellent performance on many downstream tasks and even showing better robustness than some CNNs in image scene understanding [44]. It is beneficial to solve domain transfer problems. In particular, the recently proposed SegFormer [45] and Swin transformer allow transformer models to achieve comparable performance to CNNs in terms of efficiency and computational capacity, and these models have been well explored in the field of remote sensing [46], [47].

B. Unsupervised Domain Adaptation

Compared with image classification tasks [48], domain-adaptive semantic segmentation is a more challenging task because it requires high-dimensional feature space alignment. With the same goal as fully supervised semantic segmentation, domain-adaptive semantic segmentation requires assigning a category label to each pixel in an image. However, due to the domain discrepancy problem between the training and test data, the existing models tend to have poor performance, which is essentially due to the lack of domain generalization. An intuitive theoretical idea for solving this problem is to reduce the domain shift between the images in the two domains. Therefore, generating fake target domain images using image-style transfer methods [22] such as CycleGAN and DualGAN has become a common approach. CyCADA [49] constructs a cycle consistency loss that constrains the generated fake images to have consistent semantic information with the source domain images. However, it still causes visible damage to the semantic structure of the image when the domain discrepancy is large. For the complex feature information existing in RSIs, ColorMapGAN [16] has alleviated the problem that general image-to-image translation approaches tend to change important feature information such as object shapes and textures, but this adversarial method has difficulty guaranteeing that the semantic information of an image is preserved during the transfer process. In contrast, FDA [50] provides a simple Fourier DA method that swaps low-frequency spectra to reduce the data distribution shift between the source and target domains while better preserving semantic information. In addition, the AT approach has been proven to be very effective in aligning the feature spaces of the two domains [51]. AdaptSegNet uses a multilevel output for AT to improve the transfer performance of the model by aligning the corresponding feature distributions. Compared with general binary discriminators, the discriminator of FADA [7] has more channel layers, allowing the model to incorporate category information; thus, the model can align more fine-grained category features. In particular, to address the resolution discrepancies between RSIs, Deng et al. [14] aligned the scale variation features that are specific to remote sensing objects. Recently, many state-of-the-art (SOTA) methods have indicated that ST-based methods can attain better performance in reducing domain discrepancies than AT-based methods [52], [53], [54], [55]. The key to ST approaches is filtering out high-quality pseudolabels for the target domain. CBST [52] provides a class-balance-based scheme for capturing pseudolabels. It is beneficial for the model to learn hard samples based on a certain pseudolabel selection proportion, but this approach may increase the negative impact of incorrect labels on the model. IAST [53] uses an instance-adaptive pseudolabel selection scheme but does not consider the impact of pseudolabel noise. Pseudolabel denoising is important for improving the performance of ST methods. ProDA [56] uses a pseudolabel correction method based on the class prototypes of categorywise feature centroids. Many studies on DA methods are based on the DeepLabV2 and DeepLabV3+ architectures. Although it is considered a fair comparison under the same baseline, transformer-based architectures have more potential to improve the domain generalization ability of the model.

C. Curriculum Learning

Curriculum learning conducts a progressive training strategy from easy to hard, similar to the learning processes of humans and animals [57]. Curriculum-based DA methods aim to establish easy-to-hard target tasks, thus guiding the domain transfer process of the model to improve the generalizability and convergence efficiency [58]. In essence, ST is a self-paced

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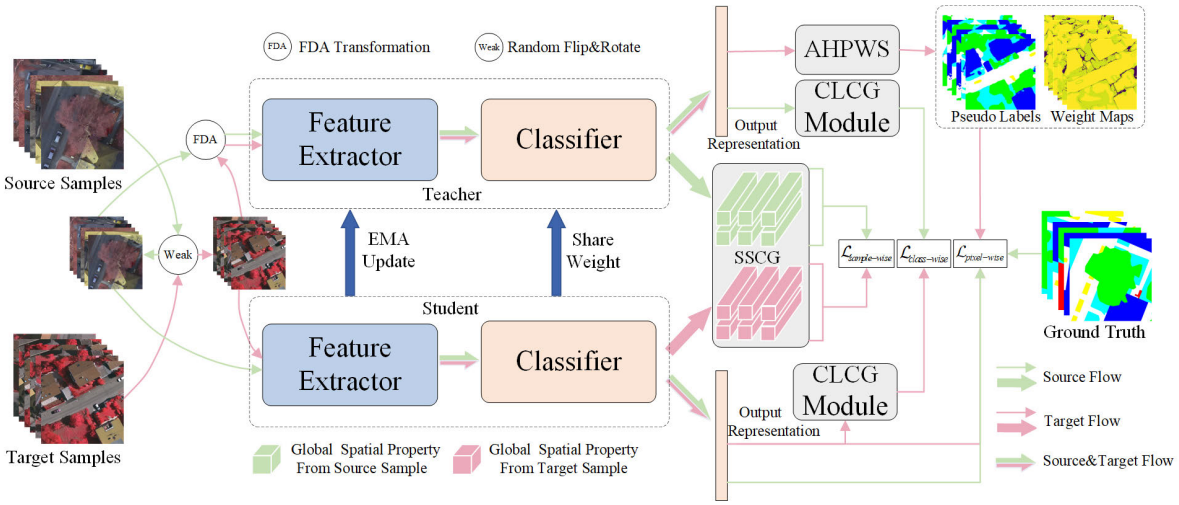


Fig. 2. Overall framework of the proposed MuGCDA. In the architecture, the teacher network is used to provide high-quality pseudolabels for the target domain while acquiring domain-robust prior features from the source domain, and its feature extractor is updated by the student network with the EMA method. The student network is used to achieve supervised learning and feature transfer process that follows curriculum ideas. SSCG mechanism guides the target domain to acquire similar sample spatial relationship properties. CLCG module acquires consistent category layout properties. AHPWS module is designed to adaptively balance pseudolabel noise with hard samples and hard class. $\mathcal{L}$ represents the multilevel loss function.

curriculum learning method [52] that follows a training process from high-confidence easy examples to low-confidence noisy examples. Domain-adaptive semantic segmentation is a worthwhile research direction for curriculum learning methods. CDAda [59] divides the target domain into easy training data and hard training samples, thus achieving smooth semantic knowledge transfer via progressive training. Zhang et al. [60] similarly divided the target domain data based on an entropy ranking principle to promote domain transfer in remote sensing scenes. Generalized curriculum learning (GCL) is not limited to operations at the input data level. A curriculum can be defined as an ordered set of training criteria [58], such as loss functions and task assignments. PyCDA [61], [62] obtains the label distribution properties of the global and squared superpixel regions according to the target domain via the ST method to guide the pseudolabel learning task. Thus, guiding the target domain to acquire the desired properties has theoretical importance in promoting the domain transfer of semantic segmentation tasks.

III. Methodology

In this section, we explain the proposed MuGCDA network for semantic segmentation tasks in detail and apply it to HR remote sensing scenes. The main goal of this work is to improve the segmentation performance of the model in an unlabeled target domain under domain discrepancy scenarios by leveraging the labeled source domain and unlabeled target domain to jointly participate in the training process. The overall DA framework is presented in Fig. 2. We formulate three-level tasks at the sample, category, and pixel levels to implement curriculum-style multilevel-guided DA. The proposed method consists of two main components: a momentum network (MN) and a segmentation network. The MN is used as a teacher network to provide the target domain with consistent prior features based on the source domain and the corresponding pseudolabels. The segmentation network works as a student network to perform supervised learning tasks while conducting feature learning in the source and target domains. Specifically, we formulate two intermediate consistent feature-guided tasks to help the semantic segmentation network better match the feature spaces of the source and target domains: the sample spatial relationship consistency guidance (SSCG) task and the CLCG task. These tasks guide the target domain to learn knowledge that is consistent with the source domain, which we term domain-robust properties. After some preliminary work is presented, the proposed MuGCDA procedure is elaborated in Section III-B. Then, in Section III-C, we introduce an AHPWS method. Compared with the methods that select pseudolabels based on confidence threshold, flexibly assigning weights to the pseudolabels increases the richness of the samples while reducing the influence of label noise.

A. Preliminaries

First, we present a general overview of unsupervised DA methods. The given source domain image is denoted as $\mathcal{X}_S = \{x_S^{(i)}\}_{i=1}^{N_S}$, and its corresponding one-hot ground truth is $\mathcal{Y}_S = \{y_S^{(i)}\}_{i=1}^{N_S}$. In general, we can simply train the semantic segmentation network g_θ on the source domain image using the multicategory cross-entropy loss

$$\mathcal{L}_S^{(i)} = -\sum_{j=1}^{H\times W}\sum_{c=1}^{C} y_S^{(i,j,c)} \log\left(\hat{y}_S^{(i,j,c)}\right) \tag{1}$$

where $\hat{y}_S^{(i,j,c)}$ is the output of the semantic segmentation network, which satisfies $\sum_c \hat{y}_S^{(i,j,c)} = 1$ and $\hat{y}_S^{(i,j,c)} \geq 0, \forall i.j,c$ after applying the softmax function. However, due to domain shift, when the network is applied to the target domain images $\mathcal{X}_T = \{x_T^{(i)}\}_{i=1}^{N_T}$ without having access to label data, the network tends to show poor performance due to its lack of domain generalization.

We use an ST-based approach to solve the domain discrepancy problem. To obtain better knowledge transfer effects,

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we use the MN h_φ as a teacher network to generate pseudolabels. A simple FDA-style transformation is applied to reduce the spectral discrepancies among the source and target images. Considering that the discrepancies between the interdomain images exist mainly in low-level feature information, which is generally captured by the feature extractor, the weights of the feature extractor in the teacher network are momentum updated based on the student network g_θ to optimize the domain transfer process. Typically, the exponential moving average (EMA) method is used

$$\varphi_t = \alpha_t \varphi_{t-1} + (1-\alpha_t)\theta_{t-1} \tag{2}$$

where α_t is updated to $(t-1/t)$ based on the current iteration step t. The upper limit is taken as 0.999.

The pseudolabels of the target domain images are acquired based on the output of the MN $\widehat{y}_T^{(i)} = h_\varphi(x_T^{(i)})$. After the pixelwise softmax function is applied, the probability vector $\widehat{y}_T^{(i,j,c)}$ satisfies $\sum_c \widehat{y}_T^{(i,j,c)} = 1$ and $\widehat{y}_T^{(i,j,c)} \geq 0$, $\forall i.j, c$. Conventional pseudolabel generation methods [1] generally select candidate samples by setting a confidence threshold τ as follows:

$$y_T^{(i,j)} = \begin{cases} c^*, & \text{if } \widehat{y}_T^{(i,j,c^*)} > \tau \\ \text{null}, & \text{otherwise.} \end{cases} \tag{3}$$

However, pseudolabels inevitably generate noise. Although this method can simply select high-quality pseudolabels and ignore hard samples by setting a high confidence threshold. Notably, the trained model is more inclined toward those easy-to-predict samples, increasing the difficulty of obtaining predictions for some hard samples and eventually causing these samples to stop participating in the training process. Based on this issue, our approach focuses more on the balance between the richness of the target domain samples and the quality of the generated pseudolabels. We also optimize the network with the multicategory cross-entropy loss to achieve knowledge transfer, and $y_T^{(i,j,c)}$ is the one-hot form of $y_T^{(i,j)}$

$$\mathcal{L}_T^{(i)} = -\sum_{j=1}^{H\times W}\sum_{c=1}^{C} y_T^{(i,j,c)} \log\left(\widehat{y}_T^{(i,j,c)}\right). \tag{4}$$

As a fine-grained pixel-level classification task, domain-adaptive semantic segmentation has considerable challenges and is a difficult problem. Therefore, we observe the similarity properties of interdomain images in terms of the sample space relationship and category layout structure, as these two properties are less affected by domain discrepancies. We use the teacher network to infer the superior property structure based on the source domain, which is an easy task. Then, the student network uses this knowledge during training, thereby allowing the target domain to learn these properties. Finally, a pixel-level domain transfer task is conducted using the ST method, which is a hard task. We formulate three-level tasks to enable the model to better match the domain-invariant feature space and realize a smooth semantic knowledge transfer process. This approach efficiently improves the stability and convergence of ST methods.

B. Multilevel-Guided Curriculum Domain Adaptation

Since the target domain cannot access the labels, we cannot directly acquire the spatial relationships and category layout properties of the target image. However, as previously mentioned, these properties are less affected by domain shift. Thus, we can infer prior properties for the target domain based on the output of the model in the source domain and ensure that the segmentation network obeys these properties to complete the knowledge transfer process. Based on this approach, we formulate a DA task with three-level objective functions

$$\mathcal{L}_{ST} = \mathcal{L}_{\text{pixel–wise}} + \lambda_{\text{sw}}\mathcal{L}_{\text{sample–wise}} + \lambda_{\text{cw}}\mathcal{L}_{\text{class–wise}} \tag{5}$$

where $\mathcal{L}_{\text{pixel–wise}}$ consists of $\mathcal{L}_S$, which uses the ground truth in the source domain, and $\mathcal{L}_T$, which uses the pseudolabels in the target domain. $\mathcal{L}_{\text{sample–wise}}$ is the SSCG loss from the source to target domains at the sample level, and $\mathcal{L}_{\text{class–wise}}$ is the CLCG loss from the source to target domains at the category level. λ_{sw} and λ_{cw} are the weights used to balance the loss.

1) Sample SSCG: First, we construct the mutual spatial relationships at the sample level. The main purpose is to transfer the domain-invariant properties of the output representation from the source domain to the target domain. Typically, given a pair of samples with indices m and n, we use the Euclidean distance to evaluate the potential relationship properties in the output representation space between the two intradomain samples

$$\psi_S^{(m,n)} = \frac{1}{WH}\left\|\sum_{j=1}^{WH}\widehat{y}_S^{(m,j)} - \sum_{j=1}^{WH}\widehat{y}_S^{(n,j)}\right\|_2 \tag{6}$$

$$\psi_T^{(m,n)} = \frac{1}{WH}\left\|\sum_{j=1}^{WH}\widehat{y}_T^{(m,j)} - \sum_{j=1}^{WH}\widehat{y}_T^{(n,j)}\right\|_2 \tag{7}$$

where $\widehat{y}_S^{(m)} = h_\varphi(x_S^m)$ is the output of the teacher network in the source domain, and $\widehat{y}_T^{(m)} = g_\varphi(x_T^m)$ is the output of the student network in the target domain. The output of the semantic segmentation network is a complex high-dimensional feature space. Thus, to capture the global spatial properties, we implement a global averaging process in the output representation space. In addition, we cannot calculate the spatial relationships of all the samples simultaneously, as this would introduce considerable computational costs and would also lead to overconstraining of the model. Therefore, when the model is trained in each iteration, we only focus on the distance relationship between pairs in mini-batch samples, and $\mathcal{X}_S^M = \{x_S^m\}_{i=1}^{M_S}$, $\mathcal{X}_T^M = \{x_T^m\}_{i=1}^{M_T}$ represents a mini-batch subset of $\mathcal{X}_S$ and $\mathcal{X}_T$, respectively.

We use the Huber loss l_δ as a penalty function to transfer potential spatial information from the teacher network to the student network, thus guiding the target domain to capture mutual spatial information from the samples. In the following function, we define the SSCG loss at the sample level, where $\psi_S^{(m,n)}$, $\psi_T^{(m,n)}$ denote the elements of the spatial relationship metric matrices ψ_S and ψ_T, respectively. λ and μ are the normalization parameters that eliminate the effects of scale

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differences in the metric matrices

$$\mathcal{L}_{\text{sample-wise}} = l_\delta\left(\frac{1}{\lambda}\psi_S, \frac{1}{\mu}\psi_T\right) \tag{8}$$

$$\lambda = \frac{1}{|\psi_S|}\sum_{x_S^m, x_S^n \in \mathcal{X}_S^M} \psi_S^{(m,n)} \tag{9}$$

$$\mu = \frac{1}{|\psi_T|}\sum_{x_T^m, x_T^n \in \mathcal{X}_T^M} \psi_T^{(m,n)}. \tag{10}$$

2) Category Layout Structure Consistency Guidance: Intuitively, the domain discrepancies reduce the discrimination ability of the model for each category in the target domain. We can easily observe that erroneous prediction regions are often caused by serious false detections with their surrounding categories. This result is expected if the transferred semantic segmentation model does not have sufficient discrimination ability for categories with small interclass variations. For instance, buildings and their surrounding impervious surfaces and low vegetation and trees show almost similar spectral characteristics. After the output of the model is normalized by the softmax function, it is difficult to discriminate the categories that have fewer feature variations based on their output probabilities, thus causing layout shifts in the whole output structure.

To alleviate this phenomenon, we consider a layout correction step at the global category level, since pixel-level optimization is strongly influenced by pseudolabel noise. It is necessary to capture the category layout properties of the target domain, which is an easier task that is more robust to noise. However, while labeled target domain images are not available, the output spaces of the two domains have similar category layout structures that are less affected by domain discrepancies. Thus, we consider that the model can learn a superior category layout structure based on the annotated source domain images, thus guiding the target domain to acquire category layout properties that are consistent with the source domain.

A module for acquiring the category layout structure based on the output representation space is shown in Fig. 3. We first obtain the region mask of each semantic category and extract the feature representation region of the corresponding category based on the mask. Then, we perform global averaging pooling (GAP) for each category region to obtain the output probability distribution of the global properties. Finally, we concatenate these layouts together to constitute a global output layout structure at the category level

$$p_S^{(i,c)} = \frac{\sum_{j=1}^{W\cdot H} \widehat{y}_S^{(i,j)} \cdot \left(\arg\max\left(\widehat{y}_S^{(i,j)}\right) = c\right)}{\sum_{j=1}^{W\cdot H}\left[\arg\max\left(\widehat{y}_S^{(i,j)}\right) = c\right]} \tag{11}$$

$$p_T^{(i,c)} = \frac{\sum_{j=1}^{W\cdot H} \widehat{y}_T^{(i,j)} \cdot \left(\arg\max\left(\widehat{y}_T^{(i,j)}\right) = c\right)}{\sum_{j=1}^{W\cdot H}\left[\arg\max\left(\widehat{y}_T^{(i,j)}\right) = c\right]} \tag{12}$$

where $\widehat{y}_S^{(i)} = \text{soft}\max(h_\varphi(x_S^i))$, $\widehat{y}_T^{(i)} = \text{soft}\max(g_\varphi(x_T^i))$, and $[\cdot]$ denotes the Iverson brackets. With this approach, we can obtain the simple category layout structure for two domain images $\mathcal{P}_S^i$, $\mathcal{P}_T^i$. Note that we use the EMA method to approximate the category layout properties of all the samples to maintain a robust output and enhance the stability of the model training process. We define the CLCG loss as follows, and we use the Kullback–Leibler divergence to minimize the objective function that allows the target domain to acquire the properties of the output space

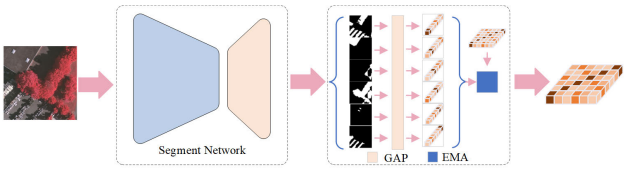


Fig. 3. CLCG module for curriculum DA.

$$\mathcal{L}_{\text{class-wise}} = \text{KL}\left(\mathcal{P}_S^i, \mathcal{P}_T^i\right). \tag{13}$$

We emphasize that capturing the output spatial relationship properties and category layout properties of the target domain is an easy task. As opposed to the model directly conducting pixel-by-pixel domain alignment, these tasks only coarsely estimate the global and regional domain-robust features of the sample. It encourages the semantic segmentation network to acquire prior domain-robust properties for the target domain, thereby providing multilevel guidance for pixel-level domain alignment.

C. AHPWS Method

The ST method uses the high-confidence prediction results as easy-to-transfer pseudolabels and then participates in the next training iteration together with the corresponding target domain images, thus progressively realizing the knowledge transfer process. However, this approach favors easy-to-transfer classes and samples that are similar to those in the source domain, while hard classes and samples that usually have lower confidence are difficult to participate in the training procedure. The intuitive idea to address this problem is to decrease the confidence threshold. However, this simple method cannot guarantee the quality of the pseudolabels and is prone to inducing more label noise. Based on this, we propose an AHPWS method. Specifically, the proposed approach introduces a weight hierarchy mechanism to balance low-confidence samples with label noise. Simultaneously, the self-adaptive weight setting method can increase the diversity of the samples to alleviate the problem that hard classes and samples are difficult to incorporate into the training process. First, we reformulate (1) and (4) as follows:

$$\begin{aligned}\mathcal{L}_{\text{pixel-wise}}^{(i)} = &-\sum_{j=1}^{H\times W}\sum_{c=1}^{C} y_S^{(i,j,c)} \log\left(\widehat{y}_S^{(i,j,c)}\right) \\ &-\sum_{j=1}^{H\times W}\sum_{c=1}^{C} \rho_T^{(i,j)} y_T^{(i,j,c)} \log\left(\widehat{y}_T^{(i,j,c)}\right).\end{aligned} \tag{14}$$

Here, the first term in the equation is the cross-entropy loss of the source domain. In the second term, $\widehat{y}_T^{(i,j)}$ is the output probability vector of the segmentation network g_θ, and $y_T^{(i,j)}$ denotes a discrete one-hot vector that is used as candidate pseudolabels for the target domain images. $\rho_T^{(i,j)}$ denotes the

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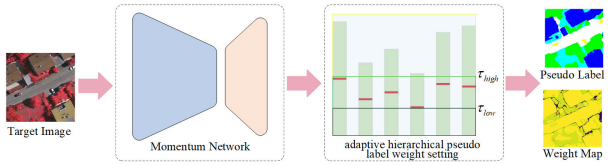


Fig. 4. AHPWS method.

pseudolabel weight corresponding to the pixel-level output. As shown in Fig. 4, we generate candidate pseudolabels and the corresponding weight masks for the target domain through an MN h_φ and propose the AHPWS method. We optimize the objective by assigning the corresponding weights to each pixel of pseudolabels, rather than directly selecting or ignoring candidate labels based on a confidence threshold. This approach is effective for balancing the sample diversity and label noise. Thus, $\rho_T^{(i,j)}$ is a critical parameter that determines the contribution of each label to model optimization. We devise the weight hierarchy mechanism to set this parameter as follows:

$$\rho_T^{(i,j)} = \begin{cases} 1, & \widehat{y}_T^{(i,j,c^*)} > \tau_{\text{high}}^{(i)} \\ \dfrac{\sum_{j=1}^{W\cdot H}\left[\tau_{\text{high}}^{(i)} > \widehat{y}_T^{(i,j,c^*)} > \tau_{\text{low}}^{(i)}\right]}{H\cdot W}, & \tau_{\text{high}}^{(i)} > \widehat{y}_T^{(i,j,c^*)} > \tau_{\text{low}}^{(i)} \\ 0, & \text{otherwise.} \end{cases} \tag{15}$$

Notably, the pseudolabel weights are determined by the parameters $\tau_{\text{low}}^{(i)}$ and $\tau_{\text{high}}^{(i)}$. Algorithm 1 presents the adaptive generation process for these two parameters. We rank the maximum probability for each category in the output of the current minibatch samples and then obtain the corresponding probability value of the category $T_{\text{thres}}^{(i,c)}$ under a certain proportion α. We adopt the highest and lowest values in $[T_{\text{thres}}^{(i,c)}]_{c=1}^{C}$ as our high and low confidence thresholds $\tau_{\text{high}}^{(i)}$ and $\tau_{\text{low}}^{(i)}$, respectively. If the category output probability is higher than the threshold $\tau_{\text{high}}^{(i)}$, the corresponding pseudolabel weight is set to 1, but if the probability is lower than the threshold $\tau_{\text{low}}^{(i)}$, the pseudolabel weight is set to 0. Crucially, the remaining candidate labels are determined by their own overall confidence degrees, with the weights set between 0 and 1. One can find that setting the threshold depending on different classes can alleviate the problem that hard classes are difficult to train, and the hierarchical weights can reduce the impact of label noise. Furthermore, the confidence thresholds $\tau_{\text{high}}^{(i)}$ and $\tau_{\text{low}}^{(i)}$ are adaptively adjusted based on the output probabilities of the current samples, so some hard samples can still be trained

$$T^{(i)} = \beta T^{(i-1)} + (1-\beta)T_{\text{thres}}^{(i)}. \tag{16}$$

Note that $\tau_{\text{high}}^{(i)}$ and $\tau_{\text{low}}^{(i)}$ depend on not only the current input samples but also the overall historical sample confidence level. We use the EMA method to adaptively update the weights of the current samples.

Algorithm 1 Procedure for Acquiring $\tau_{low}^{(i)}$ and $\tau_{high}^{(i)}$ Parameters in the Adaptive Hierarchical Pseudolabel Weight Setting Method

Input: teacher model h_φ, target domain images $\mathcal{X}_T = \{x_T^{(i)}\}_{i=1}^{N_T}$, selection proportion α, momentum parameter β

Initialization: $\alpha = 0.5, \beta = 0.9$
for $i = 1$ to N^T **do**
  Acquire the output of the teacher model h_φ based on target domain images, $P_T^{(i)} = \text{soft}\max(h_\varphi(x_T^{(i)}))$
  Generate candidate pseudolabels and corresponding maximum probability values, $P_{label}^{(i)} = \arg max(P_T^{(i)}, index = 0)$, $P_{prop}^{(i)} = \max(P_T^{(i)}, index = 0)$
  for $c = 1$ to C **do**
    Rank the maximum probability of each category, $P_{prop}^{(i,c)} = sort(P_{prop}^{(i)}[P_{label}^{(i)} == c], descending)$
    Acquire the probability value of corresponding category under proportion α, $T_{\text{thres}}^{(i,c)} = P_{prop}^{(i,c)}[length(P_{prop}^{(i,c)}) \times \alpha]$
  end for
  Get confidence threshold list for each category, $T_{\text{thres}}^{(i)} = [T_{\text{thres}}^{(i,c)}]_{c=1}^{C}$
  Update confidence thresholds based on (16), $T^{(i)} = \beta T^{(i-1)} + (1-\beta)T_{\text{thres}}^{(i)}$
  Get the highest and lowest thresholds, $\tau_{high}^{(i)} = \max(T^{(i)})$, $\tau_{low}^{(i)} = \min(T^{(i)})$
  Calculate pseudolabel weights based on (15)
end for

IV. Experimental Results and Discussion

A. Dataset Description

In this article, we use two HR remote sensing datasets published by the International Society for Photogrammetry and Remote Sensing (ISPRS) and one HR visible fine annotation dataset released by Zhongke Xingtu cup [63]. Fig. 5 shows the example images of the Potsdam dataset, Vaihingen dataset, and Gaofen-2 dataset.

1) Potsdam Dataset: The Potsdam dataset contains 38 pieces of 5-cm HR RSIs of size 6000 × 6000, which are labeled with six interpretation categories, including impervious surfaces, buildings, trees, cars, low vegetation, and clutter. The dataset consists of red, green, blue, and near-infrared (NIR) bands. We use the IR-R-G and R-G-B image modes in the experiments. Fourteen images from the Potsdam dataset are available as an online test set. The official provider has published the labels for these images and provided the corresponding accuracy verification code. In addition, among the 24 training images, we chose seven images as the validation set (IDs: 2_11, 2_12, 4_10, 5_11, 6_7, 7_8, and 7_10) and used the remaining 17 images as the training set [14], [28], [64].

2) Vaihingen Dataset: The Vaihingen dataset also includes six interpretation categories. The images have a resolution of 9 cm, and only the IR-R-G image mode is available. The dataset contains 33 very-high-resolution true orthophotos (VHR TOP) images, with 17 images designated as the official test set (IDs: 2, 4, 6, 8, 10, 12, 14, 16, 20, 22, 24, 27, 29, 31, 33, 35, and 38). Among the remaining images, we use five images as the validation set (IDs: 11, 15, 28, 30, and 34) and 11 images as the training set [2], [14], [28], [64], [65].

3) Gaofen-2 Dataset: The Gaofen-2 dataset contains 500 HR visible images of 512 × 512 size with a resolution higher than 1 m, and its labels include nine categories of road, buildings, low vegetation, trees, land, water, vehicle, impervious surfaces, and others. To conduct the Potsdam (RGB) → Gaofen-2 task, we refer to Chen et al. [24], adjusting the category labels of the Gaofen-2 dataset to be

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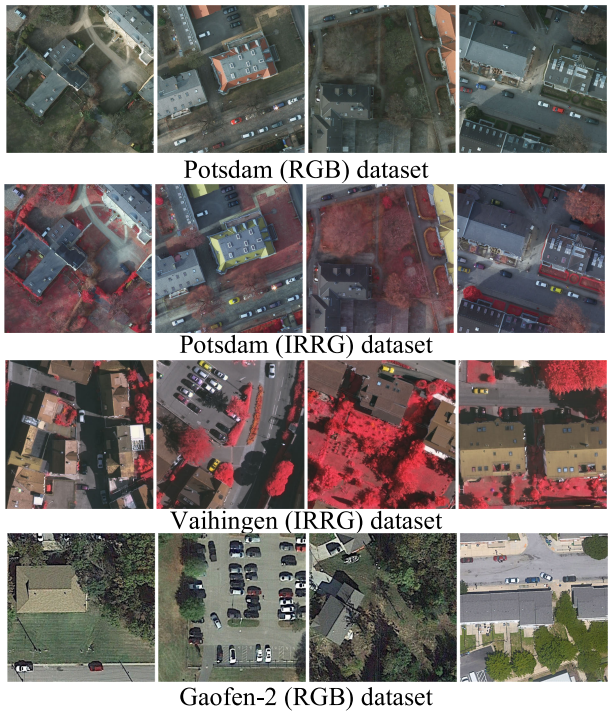


Fig. 5. Image patches of the Potsdam, Vaihingen, and Gaofen-2 datasets.

the same as the Potsdam dataset. We choose 300 images for training, 50 images as the validation set, and the remaining 150 images as the test set [24].

B. Experimental Settings and Evaluation Metrics

In this experiment, all the networks were implemented with the PyTorch framework. We used two typical semantic segmentation models, DeepLabV2 [29] and SegFormer [45], in our comparison experiments. During training, SegFormer was optimized by AdamW [66], with the momentum parameter set to 0.9 and the weight decay set to 10^{-2}. The initial learning rates of the encoder and decoder were set to 6×10^{-5} and 6×10^{-4}, respectively, and the learning rate decayed linearly as the number of iterations increased. DeepLabV2 was optimized using stochastic gradient descent (SGD) with a momentum parameter of 0.9 and a weight decay parameter of 10^{-4}. The initial learning rate was set to 2.5×10^{-4}, and the learning rate was updated by an exponential decay strategy with a power of 0.9. The whole training process included 20K iterations, and the semantic segmentation model was first iterated 1.5K times on the source domain images, which allowed the model to acquire basic semantic discrimination capability. Then, we used the proposed DA method in the next training stage. The source and target domains had batch sizes of 4. The data enhancement methods used for the semantic segmentation model included horizontal flipping, random rotations, and random cropping to a size of 512×512, which we take as weak augmentation methods. To further reduce domain discrepancies, we used the FDA method on the input data of the MN, which is a simple image transfer method.

We used the mean intersection over union (mIoU) to comprehensively evaluate the performance of the model. This metric involves calculating the IoU for each category and then determining the average value. We calculated the IoU for each category as follows, and TP, FP, and FN were calculated by the confusion matrix

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}. \tag{17}$$

C. Experimental Results and Discussion

In this section, we validate the effectiveness of the proposed method by evaluating the performance of the model on two datasets. We conduct experiments with three cross-domain remote sensing scenes: 1) cross-space scenes; 2) cross-spectral scenes; and 3) cross-space and cross-spectral scenes.

1) Results on Potsdam (IRRG) → Vaihingen (IRRG): We use the Potsdam (IRRG) dataset as the source domain data and the Vaihingen (IRRG) dataset as the target domain data for our experiments. Similar to Deng et al. [14], Liu et al. [67], and Xu et al. [68], we focus on significant objects and ignore the "clutter/background" category for both the datasets in quantitative evaluation. Thus, the results include five categories: impervious surfaces; buildings; low vegetation; trees; and cars. The Potsdam dataset mainly includes large building blocks and sparse vegetation, and due to seasonal variations, there are many barely leafy trees and areas with bare soil. In contrast, the Vaihingen dataset contains mainly small, free-standing buildings, and lush trees. These large spectral variations among the interdomain images and the large feature variances of some objects bring certain difficulties to the knowledge transfer process for the model. We compare several SOTA domain-adaptive semantic segmentation methods to verify the effectiveness of the proposed MuGCDA. Among the comparison methods, AdaptSegNet, FADA, and DecoupleNet are adversarial-based domain alignment methods, and PyCDA, IAST, DACS, and DAFormer are ST-based methods, where DecoupleNet also applies to the ST idea. To conduct a fair comparison, our method maintains the same baseline model and backbone as all the comparison methods, and consistent data processing is performed. Table I shows that all the DA models outperform the baseline model that was trained on only the source domain by 43.37%. However, the highest accuracy of 72.64% is achieved using fully supervised learning methods in the target domain, and this result is theoretically the upper limit of DA methods with DeepLabV2 as the baseline. The overall results in the table suggest that DecoupleNet achieves the best mIoU performance of 56.15% among the adversarial-based training methods, and our proposed MuGCDA improves the performance by 2.35%. DAFormer with SegFormer as the baseline is the best ST-based method among the comparison methods, with a DA mIoU of 65.81%, and our MuGCDA outperforms the best comparison model by 1.75%. In terms of the results for different categories, our MuGCDA achieves significant improvements in all the categories. The best mIoU performance is achieved for impervious surfaces, buildings, low vegetation, and trees, indicating that using multilevel consistency guidance mechanism at the sample and category levels is beneficial to facilitate the transfer process of the final semantic segmentation task in the target domain. Among the compared SOTA methods, DACS and

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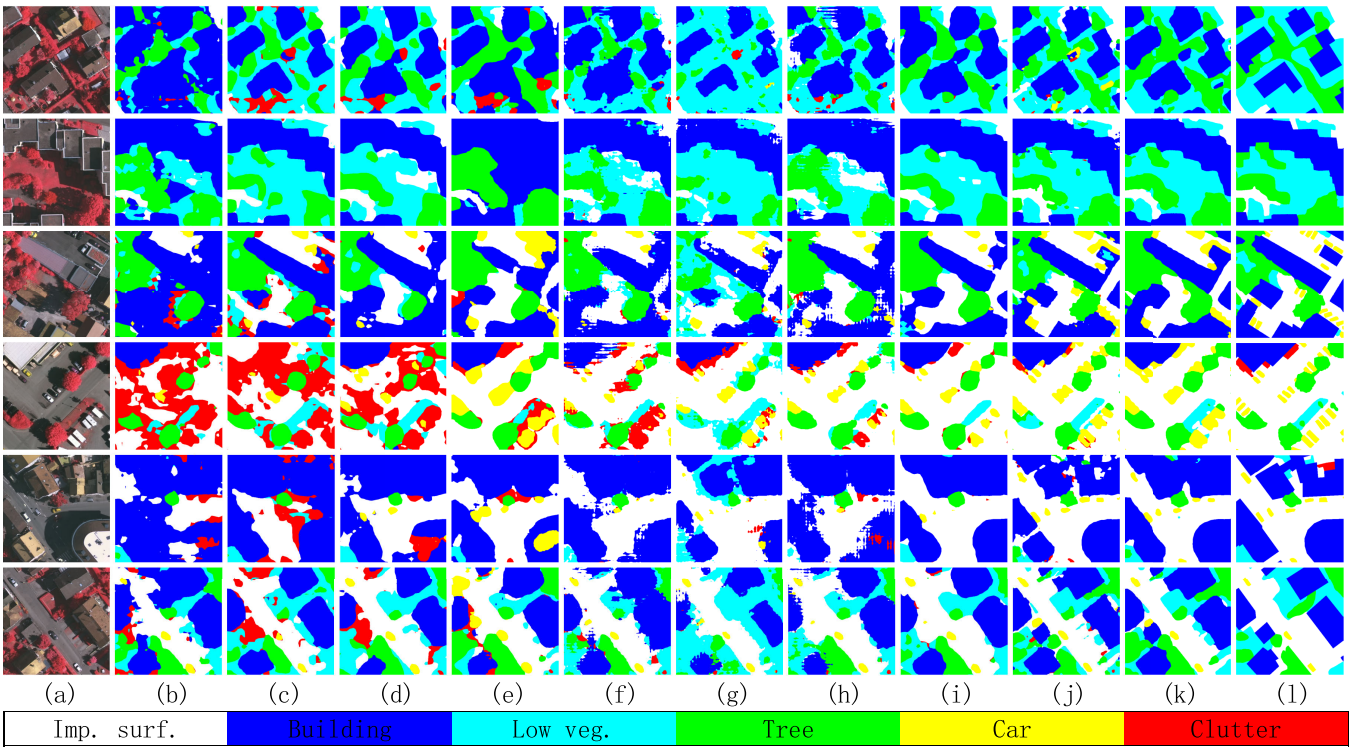


Fig. 6. Visualization results of the proposed method and the previous SOTA DA methods on the Potsdam (IRRG) → Vaihingen (IRRG) task. (a) Test images of the Vaihingen dataset. (b) Results of the source-only baseline model. (c) Results of AdaptSegNet. (d) Results of FADA. (e) Results of PyCDA. (f) Results of IAST. (g) Results of DACS. (h) Results of DecoupleNet. (i) Results of our MuGCDA with DeepLabV2 as the baseline. (j) Results of DAFormer. (k) Results of our MuGCDA with SegFormer-B2 as the baseline. (l) Ground truth.

TABLE I

QUANTITATIVE COMPARISON WITH PREVIOUS SOTA DA METHODS ON THE POTSDAM (IRRG) → VAIHINGEN (IRRG) TASK, WITH ACCURACY VALIDATIONS ON THE VAIHINGEN (IRRG) TEST SET

Network	Methods	Imp. surf.	Building	Low veg.	Tree	Car	mIoU
DeepLabV2 (ResNet101)	Source-only	45.73	58.57	34.70	57.91	19.96	43.37
	AdaptSegNet[26]	58.10	74.07	45.62	60.03	13.92	50.35
	FADA[7]	60.01	70.96	47.23	58.06	26.79	52.61
	PyCDA[61]	63.11	71.74	33.11	60.78	29.65	51.68
	IAST[53]	64.62	70.28	44.90	62.11	37.57	55.90
	DACS[54]	64.18	69.81	46.29	47.55	**42.07**	53.98
	DecoupleNet[9]	64.49	70.45	47.42	62.12	36.27	56.15
	MuGCDA(ours)	**69.02**	**76.42**	**50.24**	**62.55**	34.29	**58.50**
	Target-only	79.36	85.77	64.64	74.32	59.11	72.64
SegFormer (Mit-B2)	DAFormer[55]	76.01	81.99	51.23	**68.43**	51.40	65.81
	MuGCDA(ours)	**77.84**	**84.17**	**53.62**	67.53	**54.66**	**67.56**
	Target-only	81.68	89.92	63.83	75.80	71.65	76.58

IAST achieve better IoU performance in the car category; DACS benefits from the ClassMix strategy applied between cross-domain images, and IAST improves the quality of the pseudolabels via an instance-based pseudolabel selection method. Nevertheless, our method leverages an adaptive hierarchical setting method to determine the pseudolabel weights, thereby reducing the impact of label noise while increasing the diversity of pseudosamples. Eventually, the proposed method achieves the best segmentation performance in general. In addition, the transformer-based SegFormer architecture demonstrates more powerful domain transfer ability than the CNN architecture. Although this result is partly due to the stronger feature extraction capability of its encoder (Mit-B2), which achieves an mIoU of 76.58% through fully supervised learning in the target domain, this model also has strong advantages in its domain transfer ability, which is improved from 58.51% to 67.56%.

We present the visualized qualitative results of the proposed MuGCDA and several SOTA DA methods in Fig. 6. The white, blue, cyan, green, yellow, and red colors in the category mapping figure represent the six categories of impervious surfaces, buildings, low vegetation, trees, cars, and clutter, respectively. The results of MuGCDA are closer to the layout structure of the ground truth, and there is essentially no significant mixing. For example, because of the relatively small interclass differences, notable misdetections are observed between impervious surfaces and buildings and between low vegetation and trees in the results of AdaptSegNet and PyCDA. On the other hand,

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TABLE II

QUANTITATIVE COMPARISON WITH PREVIOUS SOTA DA METHODS ON THE VAIHINGEN (IRRG) → POTSDAM (IRRG) TASK, WITH ACCURACY VALIDATIONS ON THE POTSDAM (IRRG) TEST SET

Network	Methods	Imp. surf.	Building	Low veg.	Tree	Car	mIoU
DeepLabV2 (ResNet101)	Source-only	48.60	59.29	44.56	8.98	37.09	39.70
	AdaptSegNet[26]	**66.36**	**75.56**	52.97	25.90	46.68	53.49
	FADA[7]	61.13	62.92	48.90	33.92	41.56	49.69
	PyCDA[61]	55.33	52.83	48.02	22.37	**56.97**	47.10
	IAST[53]	62.35	68.19	53.94	39.89	43.66	53.61
	DACS[54]	42.95	63.33	37.26	35.74	46.76	45.21
	DecoupleNet[9]	59.10	59.15	55.10	42.09	46.39	52.37
	MuGCDA(ours)	60.32	65.36	**57.53**	**43.00**	52.35	**55.71**
	Target-only	82.46	90.80	72.27	74.07	79.05	79.73
SegFormer (Mit-B2)	DAFormer[55]	64.52	71.64	59.98	54.48	69.81	64.09
	MuGCDA(ours)	**70.10**	**79.13**	**62.15**	**56.39**	**72.68**	**68.09**
	Target-only	84.48	91.63	74.30	76.47	83.37	82.05

TABLE III

QUANTITATIVE COMPARISON WITH PREVIOUS SOTA DA METHODS ON THE POTSDAM (RGB) → POTSDAM (IRRG) TASK, WITH ACCURACY VALIDATION ON THE POTSDAM (IRRG) TEST SET

Network	Methods	Imp. surf.	Building	Low veg.	Tree	Car	mIoU
DeepLabV2 (ResNet101)	Source-only	72.27	75.85	10.43	66.06	66.17	58.16
	AdaptSegNet[26]	73.80	80.81	51.19	67.18	69.56	68.51
	FADA[7]	75.91	83.97	62.06	68.77	66.83	71.51
	PyCDA[61]	76.41	82.70	63.49	69.31	**73.69**	73.12
	IAST[53]	76.20	83.67	54.29	68.26	66.81	69.85
	DACS[54]	74.09	81.14	63.44	66.83	71.16	71.33
	DecoupleNet[9]	76.21	82.25	59.50	68.10	72.97	71.81
	MuGCDA(ours)	**78.05**	**86.78**	**66.84**	**69.72**	68.41	**73.96**
	Target-only	82.46	90.80	72.27	74.07	79.05	79.73
SegFormer (Mit-B2)	DAFormer[55]	77.94	86.59	71.57	**67.94**	**80.23**	**76.85**
	MuGCDA(ours)	**79.02**	**87.07**	**72.40**	67.79	80.21	**77.30**
	Target-only	84.48	91.63	74.30	76.47	83.37	82.05

although DACS, DecoupleNet, and IAST have high scores among the comparison methods, the visualization results show some unrealistic false detection noise. In contrast, the proposed method produces more regular and smoother edges for the buildings without significantly confusing this category with the surrounding impervious surfaces, demonstrating that guiding the target domain to obtain consistent category layout properties has advantageous effects.

2) Results on Vaihingen (IRRG) → Potsdam (IRRG): To further validate the generalizability of the proposed method in cross-space scenarios, we use the Vaihingen (IRRG) and Potsdam (IRRG) datasets as the source and target domains, respectively. The source domain data for the Vaihingen (IRRG) → Potsdam (IRRG) task are insufficient compared with the source domain data of the Potsdam (IRRG) → Vaihingen (IRRG) task. The Vaihingen dataset has only 11 training images with sizes of approximately 2000 × 2500, whereas the Potsdam training dataset has 17 training images with sizes of 6000 × 6000. Although insufficient data may cause more difficulties in the domain transfer process, it can verify the generalizability of the model. The quantitative results of the several SOTA methods and the proposed MuGCDA are shown in Table II. Although our method has a slightly lower accuracy than those of the compared methods for the impervious surface, building, and vehicle categories, our method achieves the best overall performance. Furthermore, our proposed MuGCDA achieves a breakthrough performance improvement with SegFormer as the segmentation network and Mit-B2 as the corresponding backbone. The visualized qualitative results of the proposed method and several compared methods are presented in Fig. 7.

3) Results on Potsdam (RGB) → Potsdam (IRRG): The different spectral bands in RSIs also lead to large domain discrepancies. Thus, we conducted a comparison experiment on the Potsdam (RGB) → Potsdam (IRRG) cross-spectral scene task to verify the effectiveness of the model. As shown in Table III, the overall accuracy of our proposed MuGCDA is higher than those of the comparison methods, which demonstrates the performance advantage of our approach in cross-spectral scenes. Cross-spectral scenes show relatively small domain discrepancies compared with the cross-space scene, with an mIoU performance of 73.96% after applying DA. This result is only 5.77% lower than the result achieved with the fully supervised approach. It is worth noting that although the mIoU of the baseline model reaches 58.51%, the baseline model achieves an mIoU of only 10.43% for the low vegetation category, which indicates that the spectral discrepancies caused by the different image bands substantially influence the ability of the model to discriminate this category. Moreover,

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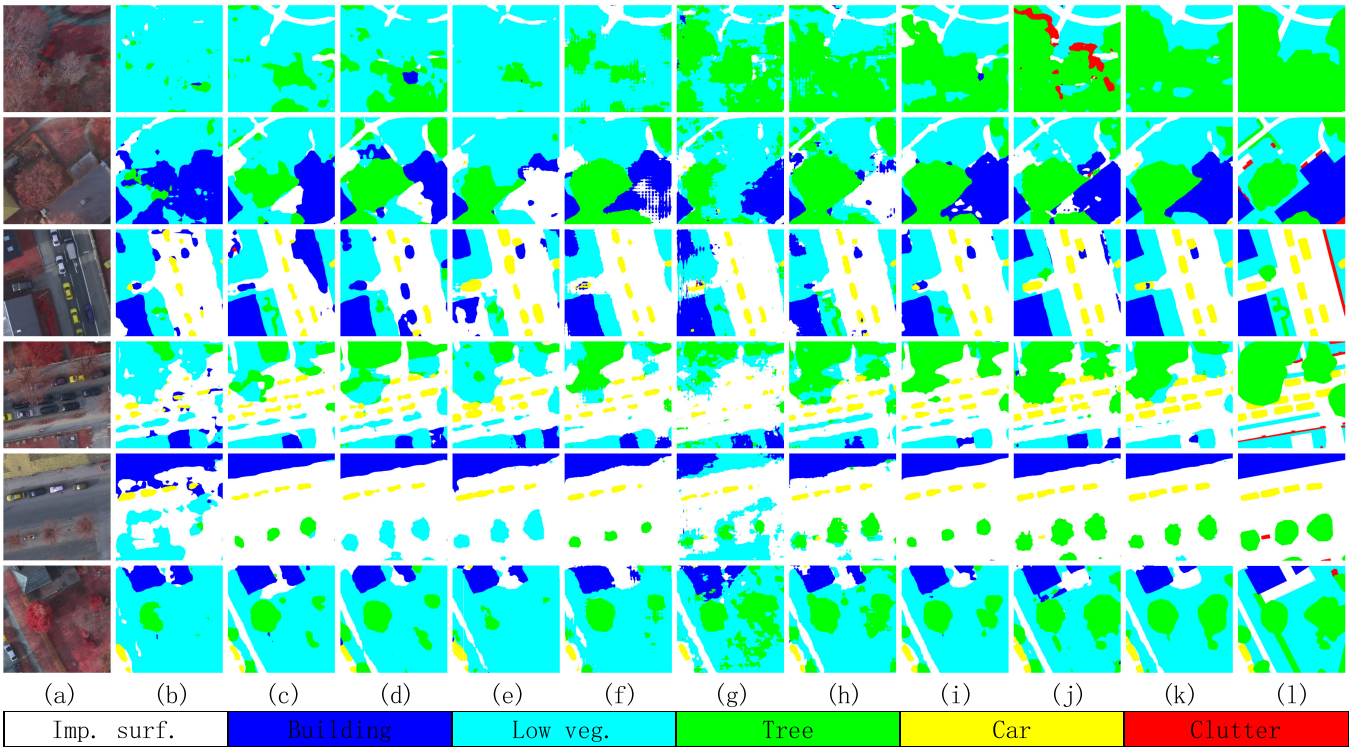


Fig. 7. Visualization results of the proposed method and the previous SOTA DA methods on the Vaihingen (IRRG) → Potsdam (IRRG) task. (a) Test images in the Potsdam dataset. (b) Results of the source-only baseline model. (c) Results of AdaptSegNet. (d) Results of FADA. (e) Results of PyCDA. (f) Results of IAST. (g) Results of DACS. (h) Results of DecoupleNet. (i) Results of our MuGCDA with DeepLabV2 as the baseline. (j) Results of DAFormer. (k) Results of our MuGCDA with SegFormer-B2 as the baseline. (l) Ground truth.

the proposed method improves the IoU performance on the low vegetation category by 56.41%, and the performance improvements for buildings, impervious surfaces, and trees are also significant, indicating that the proposed MuGCDA shows strong advantages in easy confusion categories. Furthermore, the proposed transformer-based MuGCDA further improves the performance of the model.

4) Results on Potsdam (RGB) → Vaihingen (IRRG): To further verify the effectiveness of the model in more complex and larger domain discrepancy scenes, we conduct experiments on the Potsdam (RGB) → Vaihingen (IRRG) task. Note that this task contains cross-spectral and cross-spatial scenes, which pose stronger challenges to model generalization and require DA methods to learn more representative domain-invariant feature spaces. The quantitative comparison results of the proposed method and several DA methods are presented in Table IV. The experimental results demonstrate that the proposed method is also superior in cross-spectral and cross-space scenes. Moreover, the performance of the transformer-based DA is significantly improved. It is worth mentioning that the curriculum-style PyCDA achieves the highest mIoU scores among the compared methods with the same baseline model. It is shown that the curriculum learning strategy facilitates the model to reduce the domain shift and obtain more representative domain-invariant feature spaces in large domain shift problems.

5) Results on Potsdam (RGB) → Gaofen-2: To further show the practical application value of the proposed method, we conduct experiments on the transfer task of Potsdam (RGB) → Gaofen-2. Considering that the Potsdam dataset is larger than the Gaofen-2 dataset, we perform domain transfer with Potsdam (RGB) as the source domain and the Gaofen-2 dataset as the target domain [24]. Table V shows the results of the quantitative comparison of the proposed method with several existing DA methods. Noting that the proposed method achieves the highest mIoU performance of 57.91% with SegFormer, this further demonstrates the superiority of our method for domain transfer in cross-domain scenarios of HR RSIs. In addition, our method performs better on easily confused categories, such as trees and low vegetation. Fig. 8 shows the qualitative analysis results of our method and the comparison method, and one can see that our method fits better to ground truth on each category indicating the effectiveness of the consistent guidance on category layout and the progressive transfer achieved by the curriculum ideas.

6) Ablation Study and Analysis: We next conduct ablation experiments on the Potsdam and Vaihingen validation sets. The validation set we divide is used in some methods as the final test set, so it is equally useful as a fair comparison dataset.

a) Effectiveness of each component: We conduct ablation experiments to demonstrate the effectiveness of each component. We use SegFormer with the Mit-B2 backbone as segmentation network on the Potsdam (IRRG) → Vaihingen (IRRG) task. As shown in Table VI, the performance of the conventional ST method is improved by only 1.69% over the source-only baseline model of 52.47%. Conventional ST methods adopt an offline process and simply generate pseudolabels based on confidence thresholds. FDA brought an additional 2.14% improvement. MN denotes the momentum network. The MN computes pseudolabels online during each

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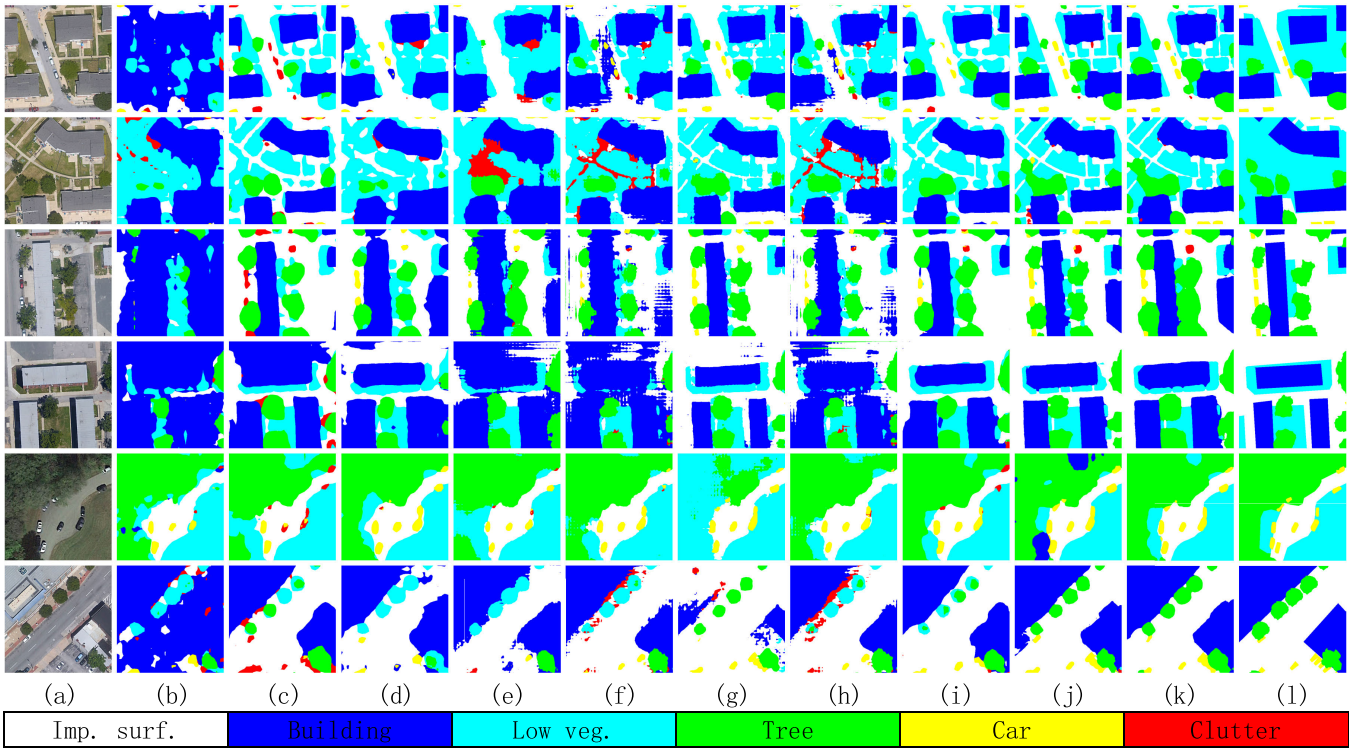


Fig. 8. Visualization results of the proposed method and the previous SOTA DA methods on the Potsdam (RGB) → Gaofen-2 (RGB) task. (a) Test images in the Gaofen-2 dataset. (b) Results of the source-only baseline model. (c) Results of AdaptSegNet. (d) Results of FADA. (e) Results of PyCDA. (f) Results of IAST. (g) Results of DACS. (h) Results of DecoupleNet. (i) Results of our MuGCDA with DeepLabV2 as the baseline. (j) Results of DAFormer. (k) Results of our MuGCDA with SegFormer-B2 as the baseline. (l) Ground truth.

TABLE IV

QUANTITATIVE COMPARISON WITH PREVIOUS SOTA DA METHODS ON THE POTSDAM (RGB) → VAIHINGEN (IRRG) TASK, WITH ACCURACY VALIDATIONS ON THE VAIHINGEN (IRRG) TEST SET

Network	Methods	Imp. surf.	Building	Low veg.	Tree	Car	mIoU
DeepLabV2 (ResNet101)	Source-only	36.68	52.15	2.46	48.04	24.52	32.77
	AdaptSegNet[26]	48.21	60.45	16.45	57.60	17.79	40.10
	FADA[7]	56.66	65.89	**36.34**	34.39	27.36	44.13
	PyCDA[61]	56.80	**72.67**	17.10	56.01	23.79	45.27
	IAST[53]	52.06	69.27	6.74	59.55	31.46	43.82
	DACS[54]	61.89	61.24	35.89	5.58	**48.24**	42.57
	DecoupleNet[9]	53.06	60.72	16.82	60.60	33.48	44.94
	MuGCDA(ours)	**63.70**	53.53	11.79	**63.59**	37.59	**46.04**
	Target-only	79.36	85.77	64.64	74.32	59.11	72.64
SegFormer (Mit-B2)	DAFormer[55]	58.85	74.20	**18.89**	62.94	46.33	52.24
	MuGCDA(ours)	**60.64**	**76.22**	15.91	**65.97**	**48.75**	**53.50**
	Target-only	81.68	89.92	63.83	75.80	71.65	76.58

iteration of training, which is found to be very effective in preserving the training stability of the model, and the mIoU performance of the model improves from 56.30% to 61.43%. Based on multilevel curriculum-based guidance mechanisms ($SSCG_{CDA}$, $CLCG_{CDA}$) to facilitate domain transfer of the semantic segmentation task, the performance is gradually and steadily improved. $SSCG_{CDA}$ indicates the SSCG mechanism for curriculum DA, and $CLCG_{CDA}$ indicates the CLCG mechanism for curriculum DA. The performance of the model is improved by 2.86% and 1.26%, respectively. We design a curriculum-style DA model with an ST paradigm. Compared with the traditional pseudolabel selection method, the proposed AHPWS method can achieve better performance. This is mainly due to the manipulation of hard samples and hard classes and the tradeoff of label noise, which finally achieves an mIoU of 67.18%. Compared with the baseline of 52.47%, the overall performance of the model is improved by 14.71%, and the experimental results demonstrate that a curriculum DA approach which guides the target domain to capture the desired properties can enhance the domain transfer effect of the model.

b) Hyperparameter sensitivity analysis: To study the influence of the hyperparameter settings on the performance of our proposed method, we analyze the sensitivity of the training loss weights. λ_{sw} and λ_{cw} represent the weights of the SSCG loss and CLCG loss, respectively. As shown in Table VII, we conducted experiments with two typical remote sensing scene tasks: the Potsdam (IRRG) → Vaihingen

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TABLE V

QUANTITATIVE COMPARISON WITH PREVIOUS SOTA DA METHODS ON THE POTSDAM (RGB) →GAOFEN-2 (RGB) TASK, WITH ACCURACY VALIDATIONS ON THE VAIHINGEN (IRRG) TEST SET

Network	Methods	Imp. surf.	Building	Low veg.	Tree	Car	mIoU
DeepLabV2 (ResNet101)	Source-only	29.03	26.19	43.17	65.95	23.27	37.52
	AdaptSegNet[26]	46.53	49.97	55.38	71.28	12.90	47.21
	FADA[7]	45.41	39.80	44.85	75.60	35.85	48.30
	PyCDA[61]	51.81	42.48	48.54	62.89	18.66	44.88
	IAST[53]	46.67	42.44	50.04	77.07	30.24	49.29
	DACS[54]	64.59	**60.64**	37.37	29.38	**44.85**	47.37
	DecoupleNet[9]	50.10	45.49	49.70	76.92	30.78	50.60
	MuGCDA(ours)	**64.06**	43.42	**56.46**	**79.56**	20.39	**52.78**
	Target-only	85.48	88.00	76.31	89.20	50.35	77.87
SegFormer (Mit-B2)	DAFormer[55]	**68.23**	36.29	49.48	70.48	**44.59**	53.81
	MuGCDA(ours)	66.46	**56.83**	**56.51**	**70.86**	38.89	**57.91**
	Target-only	89.29	91.21	78.85	90.25	62.88	82.50

TABLE VI

ABLATION EXPERIMENTS CONDUCTED WITH THE PROPOSED METHOD ON THE POTSDAM (IRRG) → VAIHINGEN (IRRG) TASK WITH SEGFORMER-B2 AS THE BASELINE

Methods	MN	$SSCG_{CDA}$	$CLCG_{CDA}$	AHPWS	mIoU	Δ
Source-only	-	-	-	-	52.47	0
Conventional ST	-	-	-	-	54.16	+1.69
FDA+Conventional ST					56.30	+2.14
+MN	✔				61.43	+5.13
+$SSCG_{CDA}$	✔	✔			64.29	+2.86
+$CLCG_{CDA}$	✔	✔	✔		65.55	+1.26
+AHPWS	✔	✔	✔	✔	67.18	+1.63
Target-only	-	-	-	-	76.10	-

(IRRG) cross-space DA scenario and the Potsdam (RGB) → Potsdam (IRRG) cross-spectral DA scenario. For the sensitivity and stability analysis, each hyperparameter was setup for five sets of data by fine-tuning. The default values of the two parameters were set to 0.1 and 0.5, respectively, for cross-validation. As can be noted, the weight λ_{sw} is relatively stable when set in the range of 0.05–0.2, while the weight λ_{cw} is stable when set in the range of 0.2–1. Furthermore, the better performance of the proposed model can be obtained when the two parameters are set to $\lambda_{sw} = 0.1$ and $\lambda_{cw} = 0.5$, respectively. Although in the cross-space scene task, the best performance is achieved when λ_{sw} and λ_{cw} are set to 0.2 and 0.5, respectively. When set to $\lambda_{sw} = 0.1$ and $\lambda_{cw} = 0.5$, the mIoU performance drops by just 0.01%. Therefore, we use $\lambda_{sw} = 0.1$ and $\lambda_{cw} = 0.5$ as the default training loss weights for the experiments.

c) Stability analysis based on different networks: We use different backbones and network architectures to conduct ablation experiments on the Vaihingen (IRRG) → Potsdam (IRRG) task, with quantitative evaluation performed on the Potsdam validation set. DeepLab and U-Net are common model architectures in many DA-related research works [14], [52], [60]. Thus, as shown in Table VIII, we also present the quantitative results achieved by the proposed DA method under these baseline models. In general, a more sophisticated model architecture implies a stronger feature extraction capability, which can improve the performance of the model. However, more powerful network architectures may lead to worse results on scenes with domain discrepancies because the extracted features may lose their generality and are prone to overfitting. DeepLabV3+ is an improved version of DeepLabV2 that has a more powerful encoder and a special atrous spatial pyramid pooling (ASPP) decoder structure. However, DeepLabV3+ with Xception65 backbone achieves a baseline (source-only) performance of 40.19%, which is 2.6% lower than that obtained by DeepLabV2. Nevertheless, the proposed MuGCDA can avoid overfitting and improve

TABLE VII

HYPERPARAMETRIC SENSITIVITY ANALYSIS OF THE PROPOSED METHOD ON THE TRAINING LOSS WEIGHTS FOR THE CROSS-SPATIAL SCENE TASK POTSDAM (IRRG) → VAIHINGEN (IRRG) AND CROSS-SPECTRAL SCENE TASK POTSDAM (RGB) → POTSDAM (IRRG). DEFAULT PARAMETER VALUES ARE MARKED WITH *

	λ_{sw}	mIoU	λ_{cw}	mIoU
Space to space	0.01	65.37	0.01	65.93
	0.05	66.86	0.2	67.09
	0.1*	67.19	0.5*	67.19
	0.2	66.90	1	67.04
	0.5	65.65	5	65.82
Spectral to spectral	0.01	75.42	0.01	75.88
	0.05	77.13	0.2	77.30
	0.1*	77.55	0.5*	77.55
	0.2	77.56	1	76.79
	0.5	76.73	5	75.21

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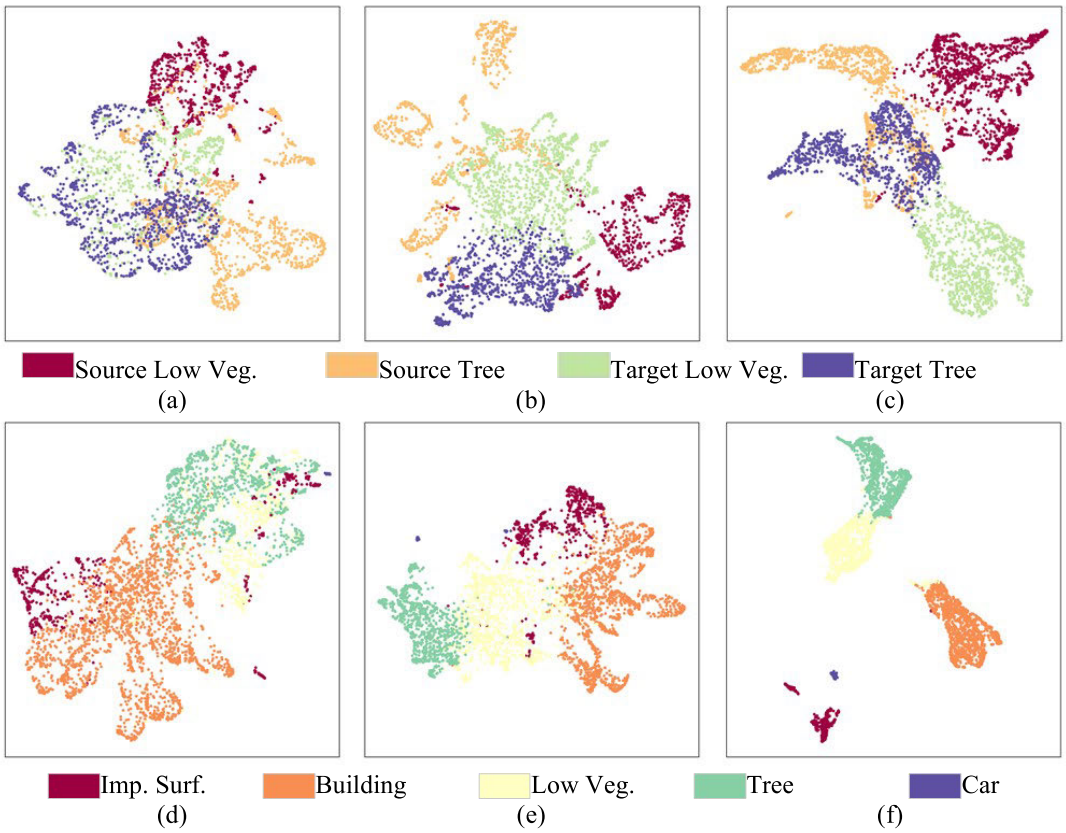


Fig. 9. Visualization of the feature space. We use the UMAP [69] reduction method to map the features to 2-D space. For a clearer illustration, in the first row, we show only the feature distributions of low vegetation and trees in the source and target domains, and in the second row, we show the feature distributions of all five classes in the target domain except for the background category. (a) and (d) Model without DA. (b) and (e) Our proposed model based on DeepLabV2. (c) and (f) Our proposed model based on SegFormer.

TABLE VIII

COMPARISON AMONG DIFFERENT VERSIONS OF THE PROPOSED METHOD WITH DIFFERENT SEGMENTATION NETWORKS. Δ DENOTES THE PERFORMANCE IMPROVEMENT OBTAINED WITH THE PROPOSED METHOD

Network	Encoder	#Params	#FLOPs	Source-only	UDA	Δ
DeepLabV2	ResNet101	42.94 M	210.58 G	42.79	57.63	↑14.84
DeepLabV3	ResNet101	45.68 M	229.99 G	42.44	57.27	↑14.83
DeepLabV3	Xception65	41.05 M	207.86 G	40.19	58.13	↑17.94
U-Net	ResNet101	51.51 M	250.07 G	44.33	60.46	↑16.13
SegFormer	Mit-B2	24.70 M	84.90 G	52.55	66.70	↑14.15

the generalizability of the model to a certain extent, and the performance after domain transfer is 0.5% better than that of DeepLabV2. Among the CNN-based semantic segmentation architectures, U-Net performs better in terms of domain generalization, and our proposed MuGCDA with the U-Net segmentation network increases the mIoU of the baseline from 44.33% to 60.46%, demonstrating a performance improvement of 16.13%. While SegFormer, which is a transformer-based model, outperforms the general CNN structure in terms of model generalization, and the baseline model reaches an mIoU of 52.55%. Overall, although the baseline capabilities of the different network architectures are affected by the domain discrepancy problem to different degrees, the proposed MuGCDA shows excellent stability in improving the DA performance.

d) Visual analysis of the feature space: To more intuitively illustrate the DA ability of the model, we visualize the learned feature space, as shown in Fig. 9. For the semantic segmentation model trained on only the source domain data, although better separation effects between categories are obtained for the source domain, the different categories in the target domain are more mixed, especially between the low vegetation and trees and between impervious surfaces and buildings. In contrast, the DA method using DeepLabV2 produces an easily separable feature space, although some dispersion remains. Furthermore, the feature space is more compact when using the transformer-based approach, and different categories of the source and target domains obtain essentially consistent feature alignment, while better separation effects are achieved between the different categories of the target domain.

V. CONCLUSION

In this article, we propose a novel MuGCDA method to address the domain discrepancy problem encountered in remote sensing semantic segmentation tasks. In contrast to

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directly performing complex pixel-level alignment on feature spaces, we first focus on domain-robust properties at the sample and category levels for guiding the target domain to acquire the corresponding consistent spatial structure. Then, we constrain the segmentation network of the target domain to use these properties when aligning the feature spaces, ultimately achieving a progressive domain transfer process from easy to hard. Although the sample-level and category-level alignment tasks are simple, roughly aligning the different prior properties of the target domain at the global level is very useful for guiding the semantic segmentation task to acquire more valuable domain-invariant features during domain transfer, which improves the generalizability of the model. Moreover, we use an online ST method to reduce the domain discrepancies of the feature space in an end-to-end manner. To balance the label noise and pseudolabel diversity for better learning hard samples and hard classes, we propose an AHPWS method. Note that the proposed MuGCDA method does not depend on special models or network structures, Thus, our model can easily be extended to other semantic segmentation networks, and MuGCDA achieves significant improvements with the application of transformer-based structures. We conduct experiments in cross-spatial scenes and cross-spectral scenes, and the results demonstrate the effectiveness and superiority of our method.

MuGCDA is a simple and effective method to achieve stable domain transfer process for various cross-domain scenes of HR RSIs. However, our method has slightly lower training efficiency than some DA methods with the same baseline and GPU, which may be caused by complex computational processes and frequent feature information interactions. We will attempt to optimize the algorithmic strategy to improve our work in future studies. In addition, MuGCDA only explores the domain-robust properties of output representations. How to guide MuGCDA to focus on domain-invariant properties at the feature level or multilevel feature level, such as scale-invariant features, to promote the model to achieve higher performance in domain discrepancy scenarios of HR RSIs is a valuable topic for future work.

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